



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Certified Lean Six Sigma White Belt (CLSSWB) Training

TGS Ref No: TGS-2025053210

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 6

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1	1 July 2026	Initial release — CLSSWB 1-day lesson plan for the Lean Six Sigma awareness course.	Dr. Alfred Ang
2	23 September 2026	Lesson plan rebuilt from the single-source content module. Schedule restructured to follow the DMAIC roadmap end to end within one 8-hour training day, with exactly one hands-on activity per DMAIC phase (5 labs) on a single continuous scenario. Content simplified to White Belt awareness depth: sigma-level and DPMO calculation, MSA, FMEA, value stream mapping, Kano analysis, weighted solution-selection matrices and SPC control limits removed. Assessment aligned to the TMS record: WA (SAQ, 30 min) plus Case Study (30 min); slide references added to the schedule.	Dr. Alfred Ang
3	23 September 2026	Lab data pack added: each activity is now a self-contained folder with its mock dataset (CSV + Excel), blank templates, a worked model answer and facilitator notes. The Lab Reference table now shows the data pack issued for each activity, and the lab steps reference the specific data and template files the learner works from.	Dr. Alfred Ang
4	23 September 2026	Labs restructured into the house ACTIVITY format: activities/ replaces labs/, one folder per activity	Dr. Alfred Ang

		named 'NN - Title', each holding the Facilitator Guide, Learner Worksheet and Checklist as DOCX and PDF plus its data pack. Lab Reference table relabelled to Activity Reference.	
5	23 September 2026	Terminology aligned: 'Lab N' now reads 'Activity N' throughout, matching the activities/ folder names and the Case Study citations. Schedule footnote clarified to state that the 480-minute total counts scheduled time excluding the lunch break.	Dr. Alfred Ang
6	23 September 2026	Completed the activity rename: the remaining learner-facing 'lab' wordings in the deck, this guide and the Lesson Plan now read 'activity'. Fixed the Lesson Plan's slide-reference lookup, which stopped matching when the deck's activity markers were renamed and had dropped the [slides N-M] references from the activity rows.	Dr. Alfred Ang

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Course Information

Course Title	Certified Lean Six Sigma White Belt (CLSSWB) Training
WSQ Course Reference	TGS-2025053210
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	1 day · 8 training hours
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on Lean Six Sigma activities using the BrewBean Cafe morning rush improvement scenario
TSC Alignment	Quality Process Control (ELE-QUA-5006-1.1)
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Describe the core concepts of quality, Lean, Six Sigma and Lean Six Sigma, and the belt roles in an improvement team.
- LO2: Explain the five phases of the DMAIC roadmap and what happens in each.
- LO3: Identify customer requirements and a clear problem statement in the Define phase.
- LO4: Describe how a process is mapped and measured, and identify the eight wastes in the Measure phase.
- LO5: Identify likely causes of a problem using 5 Whys and Fishbone analysis in the Analyze phase.
- LO6: Describe common improvement and control actions that fix a cause and hold the gain.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 30 minutes, open book.
- Case Study (CS) — applied Lean Six Sigma scenario tasks, 30 minutes, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 1 from 5:30 pm.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — Lean Six Sigma foundations and the full DMAIC roadmap, with assessment

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, course introduction, ground rules and mandatory

		digital attendance (AM) [slides 2–15]
10:00–11:00	60 min	FOUNDATIONS — What is Quality; What is Lean; What is Six Sigma; Lean vs Six Sigma vs Lean Six Sigma; the belt roles and where a White Belt contributes; the DMAIC roadmap [slides 16–26]
11:00–11:15	15 min	Tea break
11:15–12:00	45 min	DMAIC · DEFINE — Voice of the Customer; VOC to CTQ translation; problem statements; SMART goals; the project charter and scope [slides 27–39]
12:00–12:30	30 min	Hands-on Activity: Activity 1: Define — Customer Requirements and the Problem Statement [slides 40–46]
12:30–13:30	60 min	Lunch break
13:30–14:15	45 min	DMAIC · MEASURE — process mapping; SIPOC; types of data; data collection plans; check sheets; the eight wastes (DOWNTIME); value-added analysis [slides 47–58]
14:15–14:45	30 min	Hands-on Activity: Activity 2: Measure — Map the Process and Spot the Waste [slides 59–66]
14:45–15:30	45 min	DMAIC · ANALYZE — symptom vs root cause; 5 Whys; Fishbone (5M); reading a Pareto chart; common vs special cause variation [slides 67–78]
15:30–15:45	15 min	Tea break
15:45–16:15	30 min	Hands-on Activity: Activity 3: Analyze — Find the Root Cause using the 5 Whys and Fishbone tools [slides 79–86]
16:15–16:35	20 min	DMAIC · IMPROVE — generating solutions; impact/effort screening; 5S; mistake proofing (Poka-Yoke); standard work; piloting [slides 87–97]. Hands-on: Activity 4: Improve — Choose and Pilot a Countermeasure [slides 98–105]
16:35–16:55	20 min	DMAIC · CONTROL — the control plan; visual management; SOPs; team huddles; handover [slides

		106–114]. Hands-on: Activity 5: Control — Hold the Gain and Hand Over [slides 115–122]
16:55–17:30	35 min	Course recap, revision and Briefing for Assessment [slides 123–133]
17:30–18:00	30 min	Written Assessment (WA) — Short-Answer Questions (SAQ), 30 minutes, open book
18:00–18:30	30 min	Case Study (CS) — applied Lean Six Sigma scenario tasks, 30 minutes, open book. PM digital attendance

Total training time: 480 minutes (8 hours) — scheduled time from 9:30 to 18:30, excluding the 1-hour lunch break.

Activity Reference (aligned to the DMAIC phases)

DMAIC phase / Topic	Weighting	Activities	Data pack issued to learners
FOUNDATIONS: Lean Six Sigma Foundations	20%		-
DEFINE: Define — Understand the Problem	20%	Activity 1	Raw Voice of the Customer feedback (+2 blank templates)
MEASURE: Measure — See What Is Really Happening	20%	Activity 2	Morning rush observation log; Process step timings; DOWNTIME waste tally (partially completed) (+3 blank templates)
ANALYZE: Analyze — Find the Cause	20%	Activity 3	Delay reason Pareto data; Candidate causes and their evidence (+3 blank templates)
IMPROVE: Improve — Fix the Cause	10%	Activity 4	Countermeasure options with impact, effort and cost; Pilot week results (+3 blank templates)
CONTROL: Control — Hold the Gain	10%	Activity 5	Post-improvement monitoring — four weeks; Control plan — worked example (one row completed) (+3 blank templates)

Every activity is a self-contained folder holding the Facilitator Guide, Learner Worksheet and Checklist (DOCX + PDF), its mock data (CSV and Excel), blank templates and a worked model answer. The datasets are internally consistent across the five activities, so the BrewBean Cafe story reconciles from Define to Control.